WEEK 4 · DAY 1 · RULE BOOK

Scoring a Detector + Quiz 2

Follow it step by step. Every common error and its fix is here.

Your goal today: be the referee. Score every alarm from Week 3 with **one** scorer and **one** set of rules, then turn false alarms into engineer hours. **14 green ✅ ticks** to collect. Time: about 50 minutes, then Quiz 2. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open `W4D1-exercise` in Colab → **File** → **Save a copy in Drive**.
- ☐ Run **STEP 0**. Wait for ✅ `Ready!`
- ☐ Run every ▶️ cell in order. Each ✍️ cell depends on the cells above it.

The 4 scoring rules (write them down BEFORE you score)

#	Rule	Example
1	One flag anywhere inside an event's start→end (±1 time step) = **caught**	One flagged hour at 09:00 on 18 Aug catches EV01
2	Many flags inside one event = still **one** catch	EV02 flagged on both its hours = 1 catch
3	A flag outside every event = **one false alarm**	Every eclipse-hour flag = a 2 a.m. page for nothing
4	A long event is still **one** event	Flag 1 of EV05's 233 days → caught once, not 233 times

The 3 numbers

Number	Formula	Plain question	Use it?
Recall	caught ÷ real events	Did we miss anything?	✅ always
Precision	caught ÷ (caught + false alarms)	When it pages, is it real?	✅ always
Accuracy	right answers ÷ all readings	–	❌ **never** for rare events (doing nothing scores 99.87%)

The exercise, step by step

Task	What you do	The tool you need
A1	Accuracy of a detector that never fires	`1 - n_events / n_readings`
A2	Agree two scoring rules	`"yes"` / `"no"` and a number
A3	Musa the mai-guard: recall and precision	`9 / 10` and `9 / 200`
B1	Score the 82% state-of-charge floor	`SOC_FLOOR = 82`
B2	Score the control chart, all hours	`K = 3`
B3	Same chart, sunlit hours only	`eclipse_flag == 0`
B4	3σ on the amplifier residual	`N_SIGMA = 3`
B5	CUSUM on the residual	`H = 5`
B6	The event every detector missed	read the report card
B7	Your pick for the bus-power console	1, 2 or 3
C1	False alarms → hours a year	per season × 2 × 20 ÷ 60
C2	Precision of the naive rain alarm	`n_real / n_flags`
C3	Is every rain-fade hour an anomaly?	`True` or `False` (no quotes)
🌶️ **Bonus**	Build a slope alarm that catches EV06	pick `RATE_LINE`, e.g. `-3.0`

The scorer (you'll see it in the notebook)

```
score_events(flags, labels, "bus_power_telemetry_hourly.csv")          # hourly files:
tol="1h" (default)
score_events(flags, labels, "twta_thermal_daily.csv", tol="1D")         # daily files:
tol="1D"
```

`flags` is a True/False Series with the same time index as the data. The source file name must match the `source_file` column of the answer key **exactly**, including `.csv`.

When things go wrong: error → fix

You see	It means	Fix
`AssertionError: ✍️ Fill in … first`	A `___` is still there	Replace it, run the cell again
`NameError: name 'labels' is not defined`	You skipped the answer-key cell	Run the 🔑 cell (`labels = load_labels()`)

You see	It means	Fix
`NameError: name 'score_events' …`	You skipped the scorer cell	Run the ▶️ scorer cell at the start of Part B
`{'events': 0, …}` / `ZeroDivisionError`	The file name doesn't match the key	Copy it exactly: `"bus_power_telemetry_hourly.csv"`
`ParserError: Expected 8 fields …`	You loaded the labels with plain `pd.read_csv`	Use `load_labels()` : it repairs the stray comma
`ValueError: Invalid frequency: H`	Old code with capital `"H"`	Use lowercase `"h"` (and `"1h"` for tolerance)
`False alarms` look huge	You scored eclipse hours as faults	Add `& (bus["eclipse_flag"] == 0)`
Everything is weird	–	**Runtime → Restart session and run all**

Rules of thumb you'll keep forever

1. **Write the rules first**, then compute. Never change them after seeing the score.
2. **Never report accuracy** for rare events. Report recall **and** precision.
3. **Same scorer, same period** for every detector you compare.
4. **Condition on known causes** (eclipse, rain) instead of widening the band.
5. **Say false alarms in hours.** That's the number a manager can act on.
6. **Decide which number leads** before you score: recall for "a miss could cost the satellite", precision for "too many pages and people stop reading them".

🏁 **Finished?** Run `scoreboard()` . Then try the 🌶️ slope alarm, and get ready for **Quiz 2** (phones down, no AI, 20 minutes).